Project Report Template

Advanced Analytics and Applications

Team 00

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2026-04-01

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1 Introduction

This report serves as a template for your team project. It is built using Quarto, which allows you to integrate your Python or R analysis directly into your writing.

By using this template, you ensure that your formatting (margins, line spacing, and fonts) matches the required standards for submission. The left margin is set to 4cm to allow for comments, while the other margins are 2.5cm. Line spacing is set to 1.5, and the main font is Times New Roman.

2 Data Analysis Workflow

In this section, we demonstrate how to include analysis results in your report. You can either write code directly in this `.qmd` file or reference results from a separate prototyping notebook.

2.1 Prototyping in Notebooks

You can work on your analysis in the `notebooks/prototyping.ipynb` file. When you are ready to include a result, you can use Quarto's `embed` shortcode. This is a powerful feature that allows you to maintain a clean report while keeping your extensive prototyping code in separate notebooks.

To do this, you must label the specific cell in your notebook using `#| label: fig-some-name` (and optional `#| fig-cap: ...`). Then, you can embed it here like this:

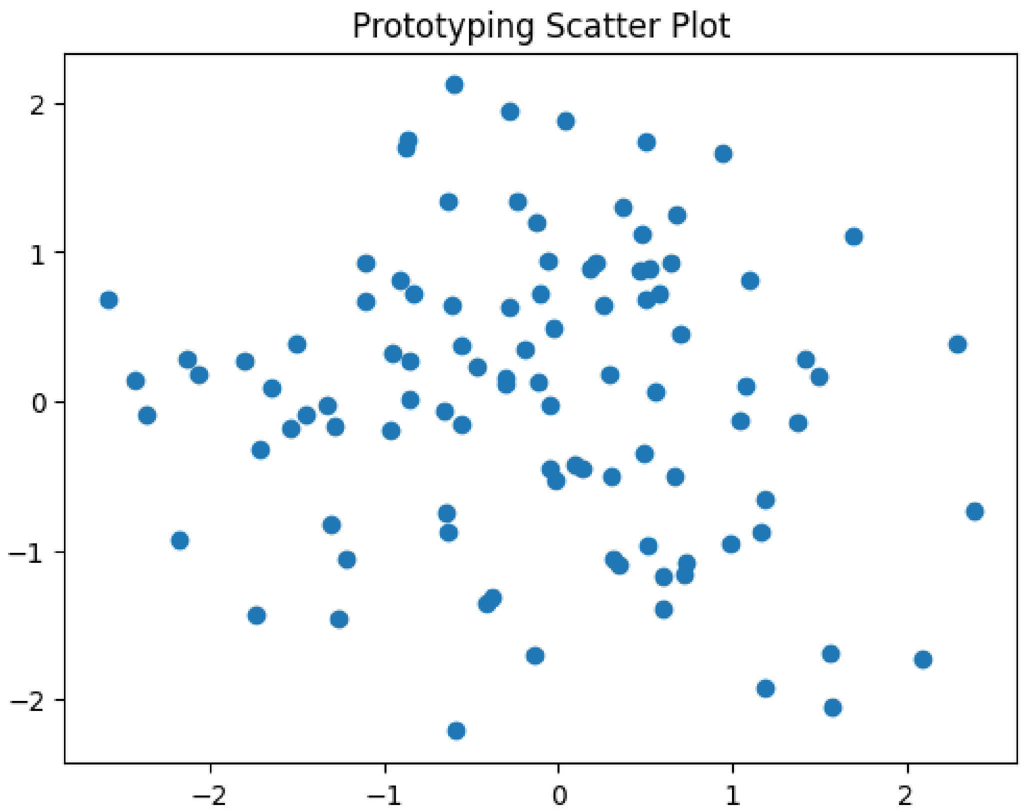

Figure 1: A plot generated in a prototyping notebook and embedded in the report.

Note that this requires your `.ipynb` notebooks to follow Quarto's cell-metadata syntax.

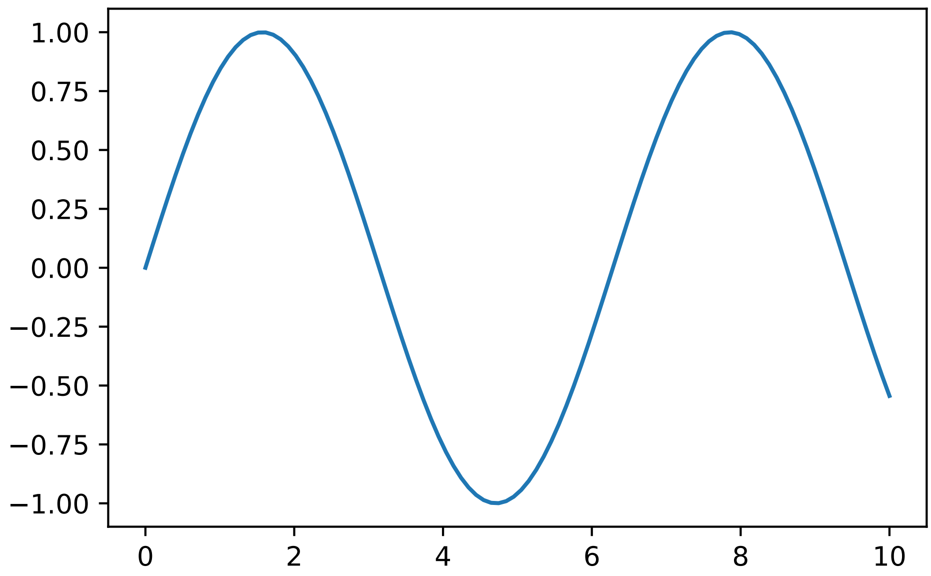

Figure 2: A sample plot generated directly in the report.

In the PDF version, the code above is hidden by default to keep the report concise and focused on the results. In the HTML version, the code can be expanded using the "Code" button.

3 Citations and References

You can easily cite your sources using BibLaTeX. For example, you can cite the Quarto documentation (Team, 2024) or a textbook like "R for Data Science" (Wickham et al., 2023). The bibliography will be automatically generated at the end of the document.

4 Conclusion

The conclusion should summarize your findings and suggest future work.

References

Team, Q. (2024). Quarto: An open-source scientific and technical publishing system. *Journal of Modern Data Science*. https://quarto.org

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science*. O'Reilly Media.